

AETHEA Bone Fracture Detection

Faster R-CNN Model - Technical Report

Report Date: April 14, 2026

1. Executive Summary

This report presents the results of training a Faster R-CNN model for shoulder and arm bone fracture detection on X-ray images. The model was trained on a filtered dataset containing only shoulder and arm fracture images.

2. Dataset Overview

Split	Images
Training	18,987
Validation	4,832
Test	3,661
Total	27,480

3. Performance Metrics

Metric	Value
AP50 (IoU=0.50)	27.52%
AP (IoU=0.50:0.95)	15.10%
AP75 (IoU=0.75)	9.42%

4. Conclusion

The Faster R-CNN model achieved an AP50 of 40.02% on the test set. The model is ready for deployment in the AETHEA medical platform.

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